IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<FIRST NAME; LAST NAME>
<DATE>

Instructions and guidelines (read carefully)

Instructions

1. Insert your name as it appears on your Online Campus learner profile in the file name. Save the file as: **First Name Last Name M10U4 Capstone Project** – **e.g., Lilly Smith M10U4 Capstone Project**.
2. Write all your answers and upload your screenshots in this PowerPoint presentation in the slides provided.
3. Convert your final presentation into **.pdf** format and submit this **PDF document**. No other file types will be accepted.

Note: Please ensure that you have checked your course calendar for the due date for this assignment.

Guidelines

Note: If you do not have a desktop version, follow the instructions from the previous lab to access PowerPoint online.

1. Complete the slides with relevant details (you are free to add additional slides, charts, and tables).
2. Add any additional unused charts or tables to the appendix that you think are relevant to your analysis.

Plagiarism declaration

1. I understand that plagiarism is to use another’s work and pretend that it is one’s own.

2. I have not allowed, and will not allow, anyone to copy my work with the intention of submitting it as their own.

3. I acknowledge that copying someone else’s assignment (or part of it) is wrong and declare that my assignments are my own work.

Plagiarism cases will be investigated in line with the Terms and Conditions for learners.

Outline

- Executive summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive summary

- Summary of methodologies
- Summary of all results

Introduction

- Project background and context
- Problems you want to solve

Section 1

Methodology

Methodology

Executive summary

- Data collection methodology
  - Describe how the data was collected.
- Perform data wrangling.
  - Describe how the data was processed.
- Perform exploratory data analysis (EDA) using visualization and SQL.
- Perform interactive visual analytics using Folium and Plotly Dash.
- Perform predictive analysis using classification models.
  - How to build, tune, and evaluate classification models

Data collection

- Describe how datasets were collected.
- Present your data collection process using key phrases and flowcharts.

Data collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts.

- Add the GitHub URL of the completed SpaceX API calls notebook (include completed code cell and outcome cell) as an external reference.

Place your flowchart of SpaceX API calls here.

Data collection – Scraping

- Present your web scraping process using key phrases and flowcharts.
- Add the GitHub URL of the completed web scraping notebook as an external reference.

Place your flowchart of web scraping here.

Data wrangling

- Describe how the data was processed.
- Present your data wrangling process using key phrases and flowcharts.
- Add the GitHub URL of your completed data wrangling notebooks as an external reference.

EDA with data visualization

- Summarize the charts plotted and why you used those charts.
- Add the GitHub URL of your completed EDA with data visualization notebook as an external reference.

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed.
- Add the GitHub URL of your completed EDA with SQL notebook as an external reference.

Build an interactive map with Folium

- Summarize the map objects (markers, circles, lines, etc.) you created and added to a Folium map.
- Explain why you added those objects.
- Add the GitHub URL of your completed interactive map with Folium as an external reference.

Build a dashboard with Plotly Dash

- Summarize the plots and interactions you added to a dashboard.
- Explain why you added those plots and interactions.
- Add the GitHub URL of your completed Plotly Dash lab as an external reference.

Predictive analysis (classification)

- Summarize how you built, evaluated, improved, and found the best-performing classification model.
- Present your model development process using key phrases and flowcharts.
- Add the GitHub URL of your completed predictive analysis lab as an external reference.

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

Flight number vs. launch site

- Show a scatter plot of flight number vs. launch site.

- Show the screenshot of the scatter plot with explanations.

Payload vs. launch site

- Show a scatter plot of payload vs. launch site.

- Show the screenshot of the scatter plot with explanations.

Success rate vs. orbit type

- Show a bar chart for the success rate of each orbit type.

- Show the screenshot of the scatter plot with explanations.

Flight number vs. orbit type

- Show a scatter plot of flight number vs. orbit type.

- Show the screenshot of the scatter plot with explanations.

Payload vs. orbit type

- Show a scatter plot of payload vs. orbit type.

- Show the screenshot of the scatter plot with explanations.

Launch success yearly trend

- Show a line chart of the yearly average success rate.

- Show the screenshot of the line chart with explanations.

All launch site names

- Find the names of the unique launch sites.
- Present your query result with a short explanation here.

Launch site names begin with “KSC”

- Find 5 records where launch sites’ names start with “KSC”.
- Present your query result with a short explanation here.

Total payload mass

- Calculate the total payload carried by boosters from NASA.
- Present your query result with a short explanation here.

Average payload mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1.
- Present your query result with a short explanation here.

First successful ground landing date

- Find the dates of the first successful landing outcome on the drone ship.
- Present your query result with a short explanation here.

Successful drone ship landing with payload between 4,000 and 6,000

- List the names of boosters that have successfully landed on a drone ship and had a payload mass greater than 4,000 but less than 6,000.
- Present your query result with a short explanation here.

Total number of successful and failed mission outcomes

- Calculate the total number of successful and failed mission outcomes.
- Present your query result with a short explanation here.

Boosters carried maximum payload

- List the names of the boosters that have carried the maximum payload mass.
- Present your query result with a short explanation here.

2015 launch records

- List the records that will display the month names, successful landing_outcomes in ground pad, booster versions, and launch_site for the months in the year 2017.
- Present your query result with a short explanation here.

Rank landing outcomes between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between 2010-06-04 and 2017-03-20 in descending order.
- Present your query result with a short explanation here.

Section 3

Launch Sites Proximities Analysis

<Folium map screenshot 1>

- Replace <Folium map screenshot 1> with an appropriate title.
- Explore the generated Folium map and make a proper screenshot to include all launch sites' location markers on a global map.
- Explain the important elements and findings on the screenshot.

<Folium map screenshot 2>

- Replace <Folium map screenshot 2> with an appropriate title.
- Explore the Folium map and make a proper screenshot to show the color-labeled launch outcomes on the map.
- Explain the important elements and findings on the screenshot.

<Folium map screenshot 3>

- Replace <Folium map screenshot 3> with an appropriate title.
- Explore the generated Folium map and show the screenshot of a selected launch site to its proximities, such as railway, highway, and coastline, with distance calculated and displayed.
- Explain the important elements and findings on the screenshot.

Section 4

Build a Dashboard with Plotly Dash

<Dashboard screenshot 1>

- Replace <Dashboard screenshot 1> with an appropriate title.
- Show the screenshot of the launch success count for all sites in a pie chart.
- Explain the important elements and findings on the screenshot.

<Dashboard screenshot 2>

- Replace <Dashboard screenshot 2> with an appropriate title.

- Show the screenshot of the pie chart for the launch site with the highest launch success ratio.

- Explain the important elements and findings on the screenshot.

<Dashboard screenshot 3>

- Replace <Dashboard screenshot 3> with an appropriate title.
- Show screenshots of the payload vs. launch outcome scatter plot for all sites, with different payloads selected in the range slider.
- Explain the important elements and findings on the screenshot, such as which payload range or booster version has the largest success rate.

Section 5

Predictive Analysis (Classification)

Classification accuracy

- Visualize the built model accuracy for all built classification models in a bar chart.

- Find the model with the highest classification accuracy.

Confusion matrix

- Show the confusion matrix of the best-performing model with an explanation.

Conclusions

- Point 1
- Point 2
- Point 3
- Point 4
- …

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, notebook outputs, or datasets that you may have created during this project.

Thank you!